CPE 381 Classwork Bonus: Laplace Transform

Student Name (as in Canvas): _____

A Number: _____

Multiple Choice Questions (1-25)

1. The Region of Convergence (ROC) of a Laplace transform is defined as the set of values of s for which the transform:
   (a) Has all its poles.
   (b) Has all its zeros.
   (c) Is analytic.
   (d) Converges absolutely.
2. For a right-sided signal, the ROC is always:
   (a) Left of the rightmost pole.
   (b) The entire s-plane.
   (c) Right of the rightmost pole.
   (d) A vertical strip.
3. A pole of a transfer function $H(s)$ is a value of s for which:
   (a) $H(s) = 0$
   (b) $H(s) = 1$
   (c) $H(s)$ is infinite
   (d) The ROC ends
4. The ROC of a stable LTI system's transfer function must contain:
   (a) The imaginary axis
   (b) The entire right-half plane
   (c) The entire left-half plane
   (d) The origin
5. If the Laplace Transform of a signal has poles at $s = -1$ and $s = -2$, and no zeros, the signal is of the form:
   (a) $e^{-t} + e^{-2t}$
   (b) $e^{t} + e^{2t}$
   (c) $\sin(t)$
   (d) $\delta(t)$
6. A zero of a transfer function $H(s)$ is a value of s for which:
   (a) $H(s) = 0$
   (b) $H(s)$ is infinite
   (c) The signal is zero

(d) The ROC begins

7. The ROC for a finite-duration signal that is zero for $t < 0$ is:

   (a) The entire s-plane
   (b) A right half-plane
   (c) A left half-plane
   (d) A strip, $\text{Re}\{s\} > \sigma_0$

8. If the ROC of $H(s)$ for a causal system is $\text{Re}\{s\} > -2$, the system is:

   (a) Stable
   (b) Unstable
   (c) Marginally stable
   (d) Cannot be determined from the information given

9. The number of poles in a rational Laplace transform $X(s)$ is:

   (a) Always greater than the number of zeros
   (b) Always equal to the number of zeros
   (c) Determined by the order of the denominator polynomial
   (d) Determined by the order of the numerator polynomial

10. For a two-sided signal, the ROC is generally:

    (a) A right half-plane
    (b) A left half-plane
    (c) A vertical strip
    (d) The entire s-plane

11. The impulse response of a system is $h(t) = e^{-3t}u(t)$. A zero of its transfer function $H(s)$ is at:

    (a) $s = -3$
    (b) $s = 0$
    (c) $s = \infty$
    (d) $H(s)$ has no zeros

12. If a system has a pole at $s = 2 + 3j$, another pole must exist at:

    (a) $s = 2 - 3j$ (for real systems)
    (b) $s = -2 + 3j$
    (c) $s = 3 + 2j$
    (d) No other pole is necessary

13. The ROC for the Laplace transform of the signal $x(t) = -e^{-at}u(-t)$ is:

    (a) $\text{Re}\{s\} > -a$
    (b) $\text{Re}\{s\} < -a$
    (c) $\text{Re}\{s\} > a$
    (d) $\text{Re}\{s\} < a$

14. A system with transfer function $H(s) = \frac{1}{s+5}$ is:

(a) Causal and stable
(b) Causal and unstable
(c) Non-causal and stable
(d) Non-causal and unstable

15. The final value theorem can be applied if:

(a) All poles of $sX(s)$ are in the right half-plane
(b) All poles of $sX(s)$ are in the left half-plane
(c) $sX(s)$ has no poles on the imaginary axis
(d) $X(s)$ has at least one pole at the origin

16. The Laplace transform of $\delta(t)$ has:

(a) One pole at the origin
(b) One zero at the origin
(c) No poles or zeros
(d) An ROC of the entire s-plane

17. A system with poles at $s = 0$ and $s = -1$, and a zero at $s = 1$, is:

(a) Stable and causal
(b) Unstable and causal
(c) Stable and non-causal
(d) Unstable and non-causal (for a standard ROC)

18. If the ROC is $\text{Re}\{s\} < -4$, the signal is:

(a) Right-sided
(b) Left-sided
(c) Two-sided
(d) Finite-duration

19. The Region of Convergence (ROC) of a Laplace transform is defined as the set of values of s for which the transform:

(a) Has all its poles.
(b) Has all its zeros.
(c) Is analytic.
(d) Converges absolutely.

20. The inverse Laplace transform is unique:

(a) Only if the ROC is specified
(b) Even without the ROC
(c) Only for causal signals
(d) Only for stable signals

21. A signal has Laplace transform $X(s) = \frac{1}{(s+1)(s+2)}$ with ROC: $-2 < \text{Re}\{s\} < -1$. The signal is:

(a) Causal
(b) Anti-causal

(c) Two-sided
(d) Stable

22. The step response of a system is related to its impulse response by:
    (a) Differentiation
    (b) Integration
    (c) Multiplication by s
    (d) Time-shifting
23. A system with transfer function $H(s) = \frac{s-1}{s+1}$ is known as:
    (a) An all-pass system
    (b) An all-pole system
    (c) A minimum-phase system
    (d) An integrator
24. The ROC for the sum of two Laplace transforms is:
    (a) The union of their individual ROCs
    (b) The intersection of their individual ROCs
    (c) Always the entire s-plane
    (d) The ROC of the first transform
25. A pole-zero plot completely defines a Laplace transform only if:
    (a) The step response is specified
    (b) The ROC is specified
    (c) The signal is causal
    (d) Both a and b

Small Numerical Questions (26-50)

26. For $X(s) = \frac{1}{s+5}$ with ROC $\text{Re}\{s\} > -5$, what is the corresponding signal $x(t)$?
27. Identify the pole and zero locations for $H(s) = \frac{s}{s+3}$.
28. A system has a pole at $s = -2 \pm 4j$. Is the system stable? (Assume causality)
29. Find the ROC for the Laplace transform of $x(t) = e^{2t}u(t) + e^{-3t}u(t)$.
30. What is the value of the Laplace transform $X(s) = \frac{s+1}{s+2}$ at its zero?
31. A signal is $x(t) = e^{-t}u(t) + e^{-2t}u(-t)$. What is the ROC of its Laplace transform?
32. How many poles does $X(s) = \frac{s^2+3s+2}{(s+1)(s+4)}$ have?
33. For $H(s) = \frac{1}{(s+1)(s-2)}$, what is the only possible ROC for which the system is stable?
34. If $h(t) = \delta(t) - 2e^{-t}u(t)$, what is $H(s)$ and its ROC?
35. Determine the final value of $x(t)$ if $X(s) = \frac{1}{s(s+1)}$, $\text{Re}\{s\} > 0$.
36. Find the poles of $X(s) = \frac{5}{s^2+4}$.

37. What is the ROC for the Laplace transform of the unit step function $u(t)$?
38. A system has impulse response $h(t) = e^{4t}u(t)$. Is it stable?
39. Identify the zero of $X(s) = \frac{s+5}{s^2+2s+5}$.
40. If a system's step response is $(1 - e^{-t})u(t)$, what is its pole?
41. For $X(s) = \frac{1}{(s+1)^2}$, what is the order of the pole at $s = -1$?
42. What is the ROC for the signal $x(t) = -e^{-at}u(-t)$?
43. A causal system has poles at $s = -1$ and $s = -3$. What is its ROC?
44. Find the Laplace transform of $\frac{d}{dt}(e^{-t}u(t))$ and state its ROC.
45. If $X(s) = \frac{s+2}{s^2+5s+6}$, simplify it and find the poles/zeros.
46. What is the initial value $x(0^+)$ for $X(s) = \frac{1}{s+10}$, $\text{Re}\{s\} > -10$?
47. A system has transfer function $H(s) = \frac{1}{s}$. What is its output for a unit step input?
48. Where is the pole for the signal $x(t) = u(t)$?
49. For $H(s) = \frac{s-1}{(s+2)(s+3)}$, what is the ROC for a stable and causal system?
50. If the ROC of a signal's Laplace transform is $\text{Re}\{s\} < -5$, is the signal left-sided or right-sided?

True/False Questions (51-75)

51. (T/F) The ROC of a stable system always includes the $j\Omega$-axis.
52. (T/F) The number of poles of a Laplace transform is always equal to the number of zeros.
53. (T/F) A causal system must have an ROC that is a right half-plane.
54. (T/F) A signal can be uniquely determined from its pole-zero plot alone.
55. (T/F) For a causal system, if a system has all its poles in the left half of the s-plane, it is stable.
56. (T/F) The Laplace transform of a signal always exists.
57. (T/F) A zero at the origin ($s = 0$) implies the system has a high-pass characteristic.
58. (T/F) The ROC never contains any poles.
59. (T/F) For a right-sided signal, the ROC is always to the right of the rightmost pole.
60. (T/F) The inverse Laplace transform of 1 is the unit impulse $\delta(t)$.
61. (T/F) A causal system with a pole at $s = 0$ is marginally stable.
62. (T/F) The Laplace transform of a sum of two signals is the product of their individual Laplace transforms.
63. (T/F) The final value theorem can be applied to the function $X(s) = \frac{1}{s-1}$.
64. (T/F) A double pole on the imaginary axis (e.g., $s = \pm 2j$) indicates instability.
65. (T/F) The ROC for a left-sided signal is a left half-plane.
66. (T/F) The step response of a system can be found by taking the inverse Laplace transform of $H(s)/s$.

67. (T/F) If the ROC of a system's transfer function is a strip, the system is two-sided.
68. (T/F) A pole-zero cancellation means a pole and a zero occurs at the same place in s-plane.
69. (T/F) The Laplace transform is a generalization of the Fourier transform.
70. (T/F) The initial value theorem depends on the ROC of the transform.
71. (T/F) A system with $H(s) = s$ is a differentiator.
72. (T/F) For a finite-duration signal, the ROC is always a vertical strip.
73. (T/F) The real part of a pole location determines the rate of exponential decay/growth of the corresponding time-domain mode.
74. (T/F) A system with no finite zeros is called an all-pole system.
75. (T/F) The ROC for the Laplace transform of a periodic signal is always a right half-plane.

Answer Key

Multiple Choice Answers

1. (d) Converges absolutely
2. (c) A right half-plane
3. (c) $H(s)$ is infinite
4. (a) The imaginary axis
5. (a) $e^{-t} + e^{-2t}$
6. (a) $H(s) = 0$
7. (a) The entire s-plane
8. (d) Cannot be determined from the information given
9. (c) Determined by the order of the denominator polynomial
10. (c) A vertical strip
11. (d) $H(s)$ has no zeros
12. (a) $s = 2 - 3j$ (for real systems)
13. (b) $\text{Re}\{s\} < -a$
14. (a) Causal and stable
15. (b) All poles of $sX(s)$ are in the left half-plane
16. (d) An ROC of the entire s-plane
17. (b) Unstable and causal
18. (b) Left-sided
19. (d) Converges absolutely
20. (a) Only if the ROC is specified
21. (c) Two-sided
22. (b) Integration
23. (a) An all-pass system
24. (b) The intersection of their individual ROCs
25. (d) Both a and b

Numerical Answers

26. $e^{-5t}u(t)$
27. Zero: $s = 0$, Pole: $s = -3$
28. Yes (poles have negative real parts)
29. $\text{Re}\{s\} > 2$ (the more restrictive right-half plane)
30. 0
31. $-2 < \text{Re}\{s\} < -1$ (a strip)
32. 2 (at $s = -1$ and $s = -4$)
33. $\text{Re}\{s\} < 1$ (a left-half plane, making it non-causal)
34. $H(s) = 1 - \frac{2}{s+1} = \frac{s-1}{s+1}$, ROC: $\text{Re}\{s\} > -1$
35. $\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) = 1$
36. $s = \pm 2j$
37. $\text{Re}\{s\} > 0$
38. No (exponentially grows)
39. $s = -5$
40. $s = -1$
41. 2 (double pole)
42. $\text{Re}\{s\} < -a$
43. $\text{Re}\{s\} > -1$
44. $\frac{s}{s+1}$, ROC: $\text{Re}\{s\} > -1$
45. $X(s) = \frac{s+2}{(s+2)(s+3)} = \frac{1}{s+3}$, Pole: $s = -3, -2$, Zero: -2
46. $\lim_{s\to\infty} sX(s) = 1$
47. A ramp, $tu(t)$
48. $s = 0$
49. $\text{Re}\{s\} > -2$
50. Left-sided

True/False Answers

51. True
52. False
53. True
54. False (ROC is also needed)

55. True (for causal systems)
56. False (e.g., e^{t^2})
57. True
58. True
59. True
60. True
61. True (for causal systems)
62. False (it's the sum)
63. False (pole is not in the LHP)
64. True
65. True
66. True
67. True
68. True (in state-space analysis)
69. True
70. False
71. True
72. False (it's the entire s-plane, possibly minus $s = 0$ or $s = \infty$)
73. True
74. True
75. False (e.g., ROC for a sine wave is a right half-plane, but for a left-sided periodic signal it would be a left half-plane)